



Giovanni Pettorru, Ph.D.

Contacts

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Languages

Italian: native
English: B2

EDUCATION

Secondary School Diploma | 2011–2016

ITC G.P. Chironi, Nuoro, Italia

- Economics sector, Administration, Finance and Marketing — specialisation in Business information systems.

Bachelor Degree — Computer Engineering | 2016–2020

University of Cagliari, Italy

- Final mark: 96/110
- Thesis: Implementation of a Magnetometer Based Vehicle Detection System for Smart Parking Applications.

Master Degree — Internet Engineering | 2020–2022

University of Cagliari, Italy

- Final mark: 110/110 cum laude
- Thesis: Secure and Low-Latency Communications Based on WebSocket over QUIC in Internet of Things Scenarios.

Doctor of Philosophy (Ph.D.) — Electronic and Computer Engineering | 2022–2026

University of Cagliari, Italy

- Final mark: cum laude
- Thesis: *Energy-Efficient and Trustworthy Location-Based Services for Next-Generation IoT*

Visiting Ph.D. Student | May 2024 – Sep 2024

Instituto Politécnico de Viana do Castelo, Portugal

PROFESSIONAL EXPERIENCE

Post-Doctoral Research Fellow | Apr 2026 – Present

University of Cagliari, Italy

Advanced research on drones as mobile localization and spatial reference platforms for spatial audio in immersive environments.

Tutor, teaching assistant | 2024–2026

University of Cagliari, Italy

Tutor for the *Network Security* course for Master's students in Internet Engineering.

Tutor, teaching assistant | 2025–2026

University of Cagliari, Italy

Tutor for *Cloud Networking* and *Digital Media* courses for Bachelor's students in Electronic, Computer, and Telecommunication Engineering.

Adjunct Professor | June 2025

University of Cagliari, Italy

Teaching activities for *Didactics for Telecommunication Systems* within the Initial University Training Program for Secondary School Teachers.

Freelance Worker | Nov 2025 – Apr 2026

University of Cagliari, Italy

Support for the development of an IoT-based platform for monitoring agricultural parameters.

Tutor, teaching assistant | 2021–2023

University of Cagliari, Italy

Tutor for the *Telecommunication Networks* course for Bachelor's students in Electrical, Electronic, and Computer Engineering.

Freelance Worker | Jul 2021 – Jan 2023

University of Cagliari, Italy

Support for the development of data acquisition algorithms on wireless networks and design of a monitoring system for alerting in critical urban situations.

OTHER ACTIVITIES & CERTIFICATES

Student Grant | Apr – Sep 2020

University of Cagliari, Italy

- M2M Comm — M2M Communication Standards in Smart Cities Environment

English B2 Level Certification | 2022

CLA, University of Cagliari, Italy

Training Course in European Project Design | 2023

Intellera Consulting S.p.a.

Specialized training in drafting and managing project proposals for 2021–2027 direct EU funding programs.

Certified Associate in Project Management (CAPM®) | 2026

Project Management Institute (PMI)

Certification of knowledge of PMI standards for project lifecycle management, planning, and control.

Research Mobility Grant (MGR) | 2026

Autonomous Region of Sardinia

Competitive international research mobility grant awarded following a merit-based evaluation of the research project proposal.

AWARDS

SERICS Lab2Bay | 2026

Fondazione SERICS @ INNOVIT

Selected for the acceleration program in San Francisco with the GreenShield project, addressing resource-aware IoT security.

Start Cup Sardegna | Oct 2025

University of Sassari, University of Cagliari and INNOIS

- Selected as one of the top 10 finalists in Start Cup Sardegna 2025.
- Special mention “Climate Change” awarded to the LENS project for its ability to integrate innovation and technology in the timely detection of extreme natural events.

Four-Minute-Thesis (4MT) | Dec 2025

IEEE ComSoc

Third place among the 20 finalists of the 4MT Competition by IEEE ComSoc at Globecom 2025.

AgriFuture | Jan 2025 – Jan 2026

Startup Desk by Sardegna Ricerche

Selected with AGROS among the top 10 high-tech innovative ideas for the agri-food sector.

Start Cup Sardegna | Oct 2024

University of Sassari, University of Cagliari and INNOIS

- Top 10 finalists
- Second prize (Spoke 7 of the e.INS) for the AGROS project.

Four-Minute-Thesis (4MT) | Dec 2024

IEEE ComSoc

Among the 15 finalists of the 4MT Competition by IEEE ComSoc at Globecom 2024.

Three-Minute-Thesis (3MT) | Jun 2023

IEEE ComSoc

Placed 3rd in the 1st Three-Minute Thesis (3MT) Competition by IEEE ComSoc at ICC 2023, where PhD students present their research to a non-specialist audience in three minutes.

PROFESSIONAL SERVICE & MEMBERSHIPS

Memberships

- IEEE Member (Institute of Electrical and Electronics Engineers) | 2021–Present
- IEEE Communications Society (ComSoc) | 2022–Present
- IEEE Young Professionals | 2021–Present
- IEEE Sensors Council | 2023–Present
- IEEE Internet of Things Community | 2023–Present

Program Committee (PC) Member | Nov 2023

AQ-IoT 2023: *Intelligent and Trustworthy Internet of Things Systems: Architectures and Qualities*.

Technical Programme Committee (TPC) Member

- ICCE Berlin 2026, *Special Session on Broadcasting*
- IEEE ICC 2027, *SAC — eHealth*

Web and Social Media Chair | 2026–Present

IEEE Communications Society (ComSoc) Italy Chapter.

Organizing Committee Member | Jul 2026

IEEE International Mediterranean Conference on Communications and Networking (MeditCom 2026).

Peer Reviewer for International Journals

IEEE Communications Letters, IEEE Transactions on Industrial Electronics, IEEE Internet of Things Journal, IEEE Transactions on Vehicular Technology, IEEE Sensors Journal, IEEE Transactions on Information Forensics and Security, IEEE Access, and Inventions (MDPI).

PUBLICATIONS

Journal Articles

- [R1] **A Cross-Layer Survey on Secure and Low-Latency Communications in Next-Generation IoT**; M. Martalò, G. Pettorru, L. Atzori; *IEEE Transactions on Network and Service Management* (2024)
- [R2] **Trustworthy Localization in IoT Networks: A Survey of Localization Techniques, Threats, and Mitigation**; G. Pettorru, V. Pilloni, M. Martalò; *MDPI Sensors* (2024)
- [R3] **Balancing Complexity and Performance in Convolutional Neural Network Models for QUIC Traffic Classification**; G. Pettorru, M. Flumini, M. Martalò; *MDPI Sensors* (2025)
- [R4] **A Persistent and Secure Publish-Subscriber Architecture for Low-Latency IoT Communications**; G. Pettorru, M. Martalò; *IEEE Transactions on Network and Service Management (TNSM)* (2025)
- [R5] **A multimodal RSSI-based dataset for indoor navigation in interference-prone environments**; G. Pettorru, V. Pilloni, M. Martalò, S.I. Lopes; *Future Generation Computer Systems*, Elsevier (2026)
- [R6] **Robust Range-Based Localization Approaches Leveraging Multiple Wireless Interfaces**; G. Pettorru, V. Pilloni, M. Martalò; *IEEE Transactions on Wireless Communications (TWC)* (2026)

Conference Proceedings

- [C1] **Implementation of a Magnetometer based Vehicle Detection System for Smart Parking applications**; A. Floris, R. Girau, S. Porcu, G. Pettorru, L. Atzori; Int. Smart Cities Conference (ISC2), IEEE (2020)
- [C2] **Implementation of a Multisensors Fire-Fighting Monitoring System for Forest Protection**; G. Pettorru, M. Bertolusso, M. Spanu, M. Sole, M. Anedda, D. Giusto; CSCf, IEEE (2022)
- [C3] **A Machine Learning-based Approach for Vehicular Tracking in Low Power Wide Area Networks**; M. Bertolusso, M. Spanu, G. Pettorru, M. Anedda, M. Fadda, R. Girau, M. Farina, D. Giusto; BMSB, IEEE (2022)
- [C4] **A passive Wi-Fi based monitoring system for urban flows detection**; M. Bertolusso, G. Pettorru, M. Spanu, M. Fadda, M. Sole, M. Anedda, D. Giusto; IAICT, IEEE (2022)
- [C5] **Pedestrian and vehicular tracking based on Wi-Fi sniffing: a real-world case study**; M. Bertolusso, G. Pettorru, M. Spanu, M. Fadda, M. Sole, M. Farina, M. Anedda, D.D. Giusto; FITCE, IEEE (2022)
- [C6] **An IoT-based electronic sniffing for forest fire detection**; G. Pettorru, M. Fadda, R. Girau, M. Anedda, D. Giusto; ICCE, IEEE (2023) (*Speaker*)
- [C7] **A Hybrid WiFi/Bluetooth RSS Dataset with Application to Multilateration-Based Localization**; G. Pettorru, V. Pilloni, M. Martalò; MeditCom, IEEE (2023) (*Speaker*)
- [C8] **Using Artificial Intelligence and IoT Solution for Forest Fire Prevention**; G. Pettorru, M. Fadda, R. Girau, M. Sole, M. Anedda, D. Giusto; ICNC, IEEE (2023)

- [C9] **QUIC and WebSocket for Secure and Low-Latency IoT Communications: an Experimental Analysis;** G. Pettorru, M. Martalò; ICC, IEEE (2023) (*Speaker*)
- [C10] **Obfuscating Sensor-Based Activity Recognition in eHealth Applications: Is Encryption Enough Secure?;** F. Marcello, G. Pettorru, M. Martalò, V. Pilloni; ICC, IEEE (2024)
- [C11] **Preserving Privacy in CSI-based Human Activity Recognition: a Data Obfuscation Case Study;** F. Marcello, G. Pettorru, M. Martalò, V. Pilloni; GlobeCom, IEEE (2024)
- [C12] **Assessing the Interplay between IoT Localization Accuracy and the Two-Ray Channel;** C. Huidobro, M.G. Gaitán, C. Oberli, G. Pettorru, M. Martalò, V. Pilloni; WCNC, IEEE (2025)
- [C13] **How Do Jamming Attacks Impact the Performance of RSS-Based Localization Techniques?;** G. Pettorru, G. Nurcis, M. Martalò, V. Pilloni; ICC, IEEE (2025)
- [C14] **Multilateration-Assisted Fingerprinting-Based Localization for Dynamic IoT Environments;** G. Pettorru, V. Pilloni, M. Martalò; MeditCom, IEEE (2025) (*Speaker*)
- [C15] **A Reliability Index for Position Estimation in Trustworthy Location-Based Services;** G. Pettorru, A. Coni, V. Pilloni, M. Martalò; MeditCom, IEEE (2025) (*Speaker*)
- [C16] **Evaluation of Resource-Aware HTTP/3 Proxies for Smuggling Resilience in IoT Environments;** L. Pisu, G. Pettorru, R. Regano, D. Maiorca, G. Giacinto, M. Martalò; GlobeCom, IEEE (2025) (*Speaker*)
- [C17] **FiSH: Feature-informed Similarity Hashing for Edge-based Traffic Fingerprinting;** A. Mancosu, G. Pettorru, M. Martalò; ICC, IEEE (2026) (*Speaker*)
- [C18] **Near-Real-Time Traffic Classification in O-RAN: A Layer 2 Approach at the 5G Edge;** E. Pilia, G. Pettorru, M. Anedda, M. Murroni; ICCE, IEEE (2026)
- [C19] **A Feasibility Study of Edge-Centric Digital Twin for Autonomous Agricultural Decision Support;** F. Falchi, G. Pettorru, V. Pilloni, M. Martalò; CAFE, IEEE (2026) (*Speaker*)
- [C20] **On the Feasibility and Byzantine Resilience of WiFi HaLow Fingerprinting-Based IoT Positioning;** G. Pettorru, M. Curridori, M. Martalò, V. Pilloni; WF-IoT, IEEE (2026)
- [C21] **Lightweight and Tamper-Evident: A Decentralized Certification Pipeline for the Internet of Energy;** G. Nurcis, A. Coni, G. Pettorru, V. Pilloni, M. Martalò; WF-IoT, IEEE (2026)